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Pipe coupling

Abstract

A pipe coupling for coupling a first pipe to a second pipe wherein a gap exists between a first pipe end and a second pipe end. The pipe coupling includes a tubular insert configured to fit within the gap and match an inside diameter and outside diameter of the pipes, thereby providing a through-butt joint between the insert and the pipes. The pipe coupling also includes a half-tubular first fitting section and a half-tubular second fitting section, configured to snap-lock and form a tubular shape over the insert, the first pipe end and the second pipe end. In some embodiments, the insert is integral with the fitting sections.

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Background/Summary

(1) This application is a continuation of U.S. application Ser. No. 18/191,335, filed Mar. 28, 2023, for PIPE COUPLING, which is a continuation of U.S. application Ser. No. 17/318,856, filed May 12, 2021, for PIPE COUPLING, now U.S. Pat. No. 11,624,462, which is a continuation of U.S. application Ser. No. 16/991,110, filed Aug. 12, 2020, for PIPE COUPLING, now U.S. Pat. No. 11,022,243, which is a continuation of U.S. application Ser. No. 14/837,808, filed Aug. 27, 2015, now U.S. Pat. No. 10,774,964, for PIPE COUPLING, all of which are incorporated in their entirety herein by reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The present invention relates generally to pipe couplings, and more specifically to split-sleeve pipe couplings.
2. Discussion of the Related Art
(2) Pipes are widely used for transporting liquids for various fluid-based systems, for example, water supply, irrigation, and wastewater systems. Pipes may be of various materials (e.g. copper, cast iron, PVC or ABS), and can be flexible or rigid.
(3) Pipe couplings are widely known in the art to connect two pieces of piping together. In one standard embodiment, two adjacent pipe ends are butted together and a tubular exterior sleeve is slid over the butt joint. The end of the new pipe section is then slid inside the exterior sleeve until abutting the existing pipe. Adhesive is typically used in conjunction with the sleeve to provide a watertight seal.
(4) With some types of piping, such as sewer piping and other gravity piping, any coupling or transition should maintain the inner surface profile of the pipe in order to prevent disruption of the flow due to changes in the inner profile of the pipe. Additionally, codes such as the Uniform Plumbing Code have other requirements such as requiring that fittings or couplings fit entirely into the pipe socket when using an adhesive or glue, and prohibiting fittings or connections that obstruct the flow.
(5) When replacing a section of piping, a watertight joint must be made with the existing pipe at each end of the new pipe section. Often, the existing piping is unmovable and of rigid composition. While a sleeve coupling may often be used on a first end of the new pipe section, after the sleeve coupling is installed on the first end, the second end of the new pipe section will be abutting the existing pipe end, preventing installation of the second pipe coupling. Therefore, a pipe coupling is needed that can provide the required watertight coupling of the butt joint without moving the pipe sections, and while maintaining the inner profile of the pipe.

SUMMARY OF THE INVENTION

(6) Several embodiments of the invention advantageously address the needs above as well as other needs by providing a coupling for coupling a first pipe with a second pipe, the first pipe and the second pipe each having a pipe inside diameter of a pipe interior surface and a pipe outside diameter of a pipe exterior surface, wherein a gap exists between a first pipe end and a second pipe end, comprising: a tubular insert with an insert inside diameter of an insert interior surface matching the pipe inside diameter, and an insert outside diameter of an insert exterior surface matching the pipe outside diameter, the tubular insert having a longitudinal length approximately equal to the gap, whereby the insert fits within the gap and generally abuts the first pipe end and the second pipe end: a first fitting section in a generally half-tubular shape and having a longitudinal section length: a second fitting section in a generally half-tubular shape and having the longitudinal section length, wherein the first fitting section and the second fitting section are configured to couple together to form a tubular section having an inside diameter generally equal to the pipe outside diameter, wherein the longitudinal length is greater than the gap, whereby the first pipe is coupled to the second pipe by fitting the insert within the gap such that the insert exterior surface aligns with the first pipe exterior surface and the second pipe exterior surface, and the first fitting section and the second fitting section are coupled together over the insert, whereby the insert, the first pipe end, and the second pipe end are enveloped by the tubular section formed by coupling the first fitting section to the second fitting section.

(7) In another embodiment, the invention can be characterized as a coupling for coupling a first pipe with a second pipe, the first pipe and the second pipe each having a pipe inside diameter of a pipe interior surface and a pipe outside diameter of a pipe exterior surface, wherein a gap exists between a first pipe end and a second pipe end, comprising: a first fitting section in a generally half-tubular shape and having a longitudinal section length, a wall thickness and an outside diameter, the first fitting section including a middle portion with a middle portion thickness greater than the wall thickness, the middle portion thickness having a middle portion inside diameter generally matching the pipe inside diameter; and a second fitting section of generally the same shape as the first fitting section, and configured to couple to the first fitting section to form a tubular section, wherein the longitudinal section length is greater than the gap, whereby the first fitting section and the second fitting section are coupled together over the first pipe and the second pipe, whereby the middle portions fit within the gap and the first pipe and the second pipe end are enveloped by the tubular section formed by coupling the first fitting section to the second fitting section.

(8) In a further embodiment, the invention may be characterized as a method for coupling a first pipe with a second pipe, the first pipe and the second pipe each having a pipe inside diameter and a pipe outside diameter, wherein a gap exists between a first pipe end and a second pipe end, comprising: fitting a tubular insert within the gap, wherein an insert inside diameter of an insert inside surface matches the pipe inside diameter, and an insert outside diameter of the insert outside surface matching the pipe outside diameter, the tubular insert having a length approximately equal to the gap; and coupling a first fitting section to a second fitting section, wherein the first fitting section and the second fitting section are generally half-tubular shapes and configured to form a tubular shape when coupled together, wherein the coupling of the first fitting section to the second fitting section envelops the tubular insert, the first pipe end and the second pipe end, and whereby the first pipe is coupled to the second pipe.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features and advantages of several embodiments of the present

invention will be more apparent from the following more particular description thereof, presented in conjunction with the following drawings.
(2) FIG. **1** is a front elevational view of the pipe coupling as installed in one embodiment of the present invention.
(3) FIG. **2** is a front perspective view of the pipe coupling as installed.
(4) FIG. **3** is a transverse cross-sectional view of the pipe coupling as installed.
(5) FIG. **4** is a longitudinal cross-sectional view of the pipe coupling as installed.
(6) FIG. **5** is a front perspective view of the pipe coupling.
(7) FIG. **6** is a transverse cross-sectional view of a fitting section of the pipe coupling.
(8) FIG. **7** is a longitudinal cross-sectional view of the pipe coupling as installed, in another embodiment of the present invention.
(9) FIG. **8** is a longitudinal cross-sectional view of existing pipe segments in a first stage of repair.
(10) FIG. **9** is a longitudinal cross-sectional view of existing pipe segments in a second stage of repair.
(11) FIG. **10** is a longitudinal cross-sectional view of existing pipe segments in a third stage of repair.
(12) FIG. **11** is a longitudinal cross-sectional view of existing pipe segments in a final stage of repair.
(13) Corresponding reference characters indicate corresponding components throughout the several views of the drawings. Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present invention.
DETAILED DESCRIPTION
(14) The following description is not to be taken in a limiting sense, but is made merely for the purpose of describing the general principles of exemplary embodiments. The scope of the invention should be determined with reference to the claims.
(15) Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.
(16) Furthermore, the described features, structures, or characteristics of the invention may be combined in any suitable manner in one or more embodiments. In the following description, numerous specific details are provided to provide a thorough understanding of embodiments of the invention. One skilled in the relevant art will recognize, however, that the invention can be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of the invention.
(17) Referring first to FIGS. **1** and **2**, a front elevation and a front perspective view of the pipe coupling **100**, as installed, are shown in one embodiment of the present invention. Shown are a first pipe **102**, a second pipe **104**, a pipe outside diameter **106**, a pipe inside diameter **108**, a gap **110**, an overlap **112**, a first pipe end **114**, a second pipe end **116**, a first fitting section **118**, a second fitting section **120**, an insert **122**, and a locking flange **124**, a longitudinal axis **126**, a pipe exterior surface **128**, a pipe interior surface **130**, a receiving notch **200**, and an exterior surface **618**.
(18) The tubular first pipe **102** and the tubular second pipe **104** represent existing pipe sections to be joined. The first pipe **102** and second pipe **104** each have the outside diameter **106** (indicated by

O.D.) and the inside diameter **108** (indicated by I.D.). The first pipe **102** and the second pipe **104** also have the pipe interior surface **130** and the pipe exterior surface **128**. Typically, the first pipe **102** and the second pipe **104** have the same outside diameter **106** and the same inside diameter **108**. It will be apparent to those of ordinary skill in the art that the typical application of the pipe coupling **100** will be for joining sections of the same inside and outside diameter, but the pipe coupling **100** may be altered to accommodate different pipe sizes. As shown in FIGS. **1** and **2**, the first pipe **102** and the second pipe **104** share the longitudinal axis **126**.
(19) The gap **110** between the first pipe end **114** and the second pipe end **116** is designated by the dimension ‘B’. The gap **110** dimension may vary but typically corresponds to a gap resulting from one of the pipe ends corresponding to a replacement section of pipe where the distal end of the pipe is installed using a standard sleeve pipe fitting. For pipe sections that are rigid and unmovable, inserting the pipe end into the sleeve fitting requires a new pipe section length smaller than the total length to be spanned by the new pipe section. The resulting gap **110** dimension can be determined by using the pipe dimensions and the size of the sleeve. It will be understood that the gap **110** B may vary, depending, for example, on the pipe diameter, or on whether the gap **110** is from installation of the distal end of a replacement section, or is the result of taking out a small width of existing pipe.
(20) The first pipe **102** and the second pipe **104** typically comprise the same material, for example, PVC, CPVC, or ABS. In some embodiments, the first pipe **102** and the second pipe **104** are Schedule 40, Schedule 80, or SDR 35.
(21) The pipe coupling **100** comprises the first fitting section **118**, the second fitting section **120**, and the insert **122**. In the embodiment shown, the first fitting section **118** and the second fitting section **120** are essentially the same shape. The first fitting section **118** and the second fitting section **120** are typically of the same material as the first pipe **102** and the second pipe **104**. Additionally, the first fitting section **118** and the second fitting section **120** are of a material resiliently flexible enough to allow for the snap-lock joints coupling the first fitting section **118** to the second fitting section **120**. For example, the first fitting section **118** and the second fitting section **120** may comprise PVC, CPVC or ABS. It will be understood that for other methods of coupling the first fitting section **118** to the second fitting section **120**, for example a screw-type connecting, the fitting sections may comprise a more rigid material. The insert **122** may be the same material as the first pipe **102** and the second pipe **104**, the same material as the first fitting section **118** and the second fitting section **120**, or a different material suitable for conveyance of a specific liquid and/or for forming waterproof joints between the insert **122**, the first pipe **102** and the second pipe **104**.
(22) For clarity, only the first fitting section **118** will be described, as it will be understood that the first fitting section **118** description applies equally to the second fitting section **120**.
(23) The first fitting section **118** is a generally half-tubular shape, i.e. a tube section cut in half longitudinally. The first fitting section **118** has an inside diameter generally equal to the O.D. of the pipe sections. An angle between the receiving end and the locking flange end (as shown in FIG. **6**) is greater than 180 degrees, such that when the first fitting section **118** and the second fitting section **120** and are placed over the pipes, the locking flange **124** of each fitting section overlaps and snap-locks with the receiving end of the other section, forming a tubular shape with an inside diameter generally equal to the pipe outside diameter **106**.
(24) In the embodiment shown, the first fitting section **118** and the second fitting section **120** each have a longitudinal length A+B+A, where B is the gap **110** dimension as previously described, and A is a the overlap **112** dimension on each side of the gap **110** dimension. In some embodiments, the overlap **112** dimension may be different on each side of the gap **110**.
(25) In some embodiments, the overlap **112** dimension A is a standard pipe fitting overlap dimension based on at least one of the pipe material, thickness, and diameter.
(26) The insert **122** is a tubular shape with an insert outside diameter matching the pipe outside

diameter **106** and an insert inside diameter matching the pipe inside diameter **108**. The insert also has an insert interior surface **400** and an insert exterior surface **402** (as shown in FIG. **4**). A length of the insert **122** is approximately equal to, or slightly smaller than the gap **110**, such that the insert **122** may be inserted into the gap **110**, generally abutting the first pipe end **114** and the second pipe end **116**.

(27) It will be understood by those of ordinary skill in the art that in lieu of the separate insert **122**, the pipe coupling may be comprised of two fitting sections **118**, **120** with an integral middle portion **702** forming the interior shape of the insert **122**, as shown below in FIG. **7**. The following descriptions of the pipe coupling **100** with the separate insert **122** apply equally to the integral embodiment shown in FIG. **7**.

(28) In operation, the first pipe **102** and second pipe **104** are existing, with the gap **110** B between the first pipe end **114** and the second pipe end **116**. The insert **122** is placed between the first pipe **102** and the second pipe **104**, with an insert longitudinal axis aligning with the pipe longitudinal axis **126**. A waterproof adhesive is applied between the first pipe end **114** and the insert **122** and the second pipe end **116** and the insert **122**. The waterproof adhesive provides a waterproof seal between the insert and the first pipe end, and between the insert and the second pipe end.

(29) The waterproof adhesive is applied to the exterior of the first and second pipe **104** for an approximate distance of A from each pipe end. The waterproof adhesive is also applied to the exterior of the insert **122**. The first fitting section **118** is placed around the first pipe **102**, the second pipe **104**, and the insert **122** such that the first fitting section **118** is generally centered on the insert **122**, as shown in FIG. **1**. The second fitting section **120** is then coupled to the first fitting section **118**, forming a tubular section and enveloping the first pipe end, the second pipe end, and entire insert. In the preferred embodiment, the coupling is a snap-lock type coupling, as described further below.

(30) When it is necessary to replace a section of pipe using butt joints that are glued all the way into the pipe cavity, and when the original pipe sections are rigid and non-movable, conventional sleeve fittings cannot be used on both ends of the replacement pipe. Additionally, when a sleeve fitting is used on one end of the replacement section, the replacement section must be shorter than the total replacement length so that replacement section can be aligned with the existing pipe ends, then slid into the fitting. The sliding of the fitting into the sleeve creates the gap **110** at the other end. The present invention provides the code-compliant butt joint while accommodating the gap **110** caused by the connection on the other end. Pipe replacement using the pipe coupling **100** is shown further below in FIGS. **8-11**. Additionally, the present invention maintains the inner diameter of the existing pipe, as the insert **122** is chosen to include an inside diameter to match the existing pipe. This is important for pipes such as sewage pipes, which require a smooth, generally continuous, inner surface to avoid blockage and comply with code requirements, such as UPC 316.4.1, which prohibits fittings or couplings with an enlargement, chamber, or recess with a ledge, shoulder, or reduction of pipe area that offers a reduction to flow through the drain. The fitting may be used for any compatible piping, for example, plumbing drains, vent pipes, sewer lines, and electrical conduits.

(31) Referring next to FIG. **3**, a transverse cross-sectional view of the pipe coupling **100** installed on the first pipe **102** and the second pipe **104** is shown. Shown are the second pipe **104**, the pipe outside diameter **106**, the pipe inside diameter **108**, the first fitting section **118**, the second fitting section **120**, the plurality of locking flanges **124**, the plurality of notches **200**, a plurality of teeth **302**, a plurality of receiving ends **304**, the pipe interior surface **128**, the pipe exterior surface **130**, and a waterproof adhesive **306**.

(32) As previously described, the first fitting section **118** and the second fitting section **120** are coupled together over the insert **122** and the first pipe **102** and the second pipe **104** to form the exterior pipe coupling **100**. As shown in FIG. **3**, the first fitting and the second fitting each include the locking flange **124** at one longitudinal end (the locking flange end) and the notch **200** at the

opposite longitudinal end (the receiving end **304**). The locking flange **124** includes the tooth **302** on the interior surface **616** of the fitting section. The receiving end **304** includes the notch **200** on the exterior surface **618** of the fitting. In operation, the locking flange **124** flexes outward enough for the tooth **302** of the first fitting section **118** to pass a portion of the receiving end **304** of the second fitting section **120** and be received in the notch **200** of the second fitting section **120**, forming the snap-lock coupling of the locking flange end to the receiving end **304**. The locking flange end of the second fitting section **120** is coupled to the receiving end **304** of the first fitting section **118** in the same way. The shape of the fitting section is described in more detail in FIG. **6**. As previously mentioned, the coupling **100** may also be formed with the insert **122** integral to the fitting sections **118**, **120** as shown in FIG. **7**.
(33) Referring next to FIG. **4**, a longitudinal cross-sectional view of the pipe coupling **100** installed on the first pipe **102** and the second pipe **104** is shown. Shown are the first pipe **102**, the second pipe **104**, the pipe outside diameter **106**, the pipe inside diameter **108**, the gap **110**, the plurality of overlaps **112**, the first pipe end **114**, the second pipe end **116**, the first fitting section **118**, the second fitting section **120**, the insert **122**, the pipe interior surface **128**, the insert interior surface **400**, the insert exterior surface **402**, and the waterproof adhesive **306**.
(34) As shown in FIG. **4**, the pipe coupling **100** is configured such that the insert **122** nominally matches the gap **110** B between the first pipe end **114** and the second pipe end **116**. It will be understood by those of ordinary skill in the art that the insert **122** will be slightly shorter than the gap **110** to allow for construction tolerances and gluing. In some embodiments the insert **122** is ⅙″-⅛″ shorter than the gap. As previously described, the insert inside diameter is generally the same as the inside diameter **108** of the first pipe **102** and the second pipe **104**, to provide a generally constant inner surface.
(35) As previously mentioned, the coupling **100** may also be formed with the insert **122** integral to the fitting sections **118**, **120** as shown in FIG. **7**.
(36) Referring next to FIG. **5**, a front perspective view of the pipe coupling **100** is shown. Shown are the overlap **112**, first fitting section **118**, the second fitting section **120**, the insert **122**, the plurality of locking flanges **124**, the pipe interior surface **128**, the pipe exterior surface **130**, the plurality of receiving ends **304**, the insert interior surface **402**, and an insert end surface **500**.
(37) As shown in FIG. **5**, the insert **122** is located inside the first fitting section **118** and the second fitting section **120** such that each end of the insert **122** is approximately the overlap **112** dimension A from the proximate end of the tubular section.
(38) Referring next to FIG. **6**, a cross-sectional view of an exemplary fitting section **600** is shown in one embodiment of the invention. Shown are the locking flange **124**, the notch **200**, the tooth **302**, the receiving end **304**, a fitting inside radius **602**, a fitting outside radius **604**, a fitting flange radius **606**, a receiving end distance **608**, a first locking face **610**, a second locking face **612**, the triangular notch **200**, an interior surface **616**, an exterior surface **618**, a flange shoulder **620**, a shoulder distance **622**, a third locking face **624**, a locking flange end **626**, and a locking flange end distance **628**, the angle **630**, and a locking flange notch **632**.
(39) As previously described, the first fitting section **118** and the second fitting section **120** are the same shape, herein described as the exemplary fitting section of FIG. **6**.
(40) As previously described, the fitting section **600** is the generally half-tubular shape. The fitting section **600** has the inside radius **602** and the outside radius **604**, with respect to the center **126** of the fitting section **600**. The center **126** of the fitting section **600** corresponds to the pipe longitudinal axis **126**. The fitting section ends parallel to the longitudinal axis **126** are the receiving end **304** and the locking flange end. The angle between the receiving end **304** and the locking flange end is greater than 180 degrees. A thickness of the fitting section **600** is generally equal to the difference between the inside radius **602** and the outside radius **604**.
(41) The receiving end **304** of the fitting section **600** may be rounded. The receiving end **304** includes the triangular notch **200** on the exterior surface **618** of the fitting section **600**. The notch

200 includes the second locking face **612**, which is generally perpendicular to the exterior surface **618**. The second locking face **612** has a depth, which in the embodiment shown is not more than 50% of the thickness of the fitting section **600**. The second locking face **612** is proximate to the receiving end **304**. The notch **200** is formed by linearly sloping up from the bottom of the second locking face **612** to the exterior surface **618** of the fitting section **600** in a direction away from the receiving end **304**. In one embodiment, the distance from the receiving end **304** to the second locking face **612** is approximately 7/16″. The depth of the second locking face is ⅛″, and a width of the notch **200** along the exterior surface **618** is approximately ½″.
(42) At the opposite end, the locking flange end **626**, at approximately 180 degrees from the receiving end **304** the fitting section **600** is jogged outward, forming the third locking face **624** approximately perpendicular to the interior surface **616** of the fitting section **600**, and the flange shoulder on the exterior surface **618**. The location of the third locking face **624** is configured to receive the receiving end **304** when two sections are locked together. The third locking face **624** depth approximately matches the thickness of the fitting section **600**. The outwardly jogged portion forms the locking flange notch **632** in the locking flange **124**. The locking flange notch **632** terminates in the tooth **302** comprising the first locking face **610**. The locking flange end distance **628** between the first locking face **610** and the locking flange end **626** is slightly smaller than the receiving end distance **608**, so that the tooth **302** may be received by the triangular notch **200** of the other fitting section **600**.
(43) The locking flange end **626** includes the fitting flange radius **606** that is greater than the outside diameter of the fitting section **600**. The fitting flange radius **606** allows for the fitting section **600** to maintain enough thickness when the fitting section **600** is jogged outward to form the locking flange notch **632**. The extent of the fitting section **600** having the fitting flange radius **606** is from the locking flange end **626** to the flange shoulder **620**, which is equal to the locking flange end distance **628** plus the shoulder distance **622**. In the embodiment shown, the extent of the fitting flange radius **606** is 1.75″.
(44) The cross-sectional shape of the fitting section **600** is generally the same for the entire longitudinal length of the fitting section **600**.
(45) As previously mentioned, the fitting section **600** may also be formed with the insert **122** integral with the fitting section **600** as shown in FIG. **7**.
(46) Referring next to FIG. **7**, a cross-sectional view of the pipe coupling **100** installed on the pipe is shown in a direction parallel to the longitudinal axis **126**, in another embodiment of the present invention. Shown are the first pipe **102**, the second pipe **104**, the pipe outside diameter **106**, the pipe inside diameter **108**, the gap **110**, the overlap **112**, the first pipe end **114**, the second pipe end **116**, the first fitting section **118**, the second fitting section **120**, the longitudinal axis **126**, the pipe interior surface **128**, the pipe exterior surface **130**, a wall thickness **700**, and a middle portion **702**.
(47) In another embodiment of the present invention, the insert **122** may be formed as an integral part of the first fitting section **118** and the second fitting section **120**, as shown in FIG. **7**. Each fitting section is formed, for example by injection molding, with the additional thickness of the insert **122** included in the fitting section shape. The integral design has the advantage of not requiring the insert **122** to be held in place while the butt joints are glued, and eliminating the joint on the exterior of the insert **122**. A horizontal glued joint would be formed between the insert portions at the snap-lock location.
(48) Referring next to FIGS. **8-11**, longitudinal cross-sectional views of an existing pipe with a replacement segment are shown in various stages of repair. Shown are the first pipe **102**, the second pipe **104**, the pipe coupling **100**, the gap **110**, a third pipe **800**, a pipe distance **802** and a standard pipe coupling **900**.
(49) In a first stage shown in FIG. **8**, the second pipe **104** on the left and the third pipe **800** on the right are shown as existing portions of a pipeline to be joined. The second pipe **104** and the third pipe **800** are separated by the pipe distance **802**.

(50) In a second stage shown in FIG. **9**, the standard pipe fitting **900** has been coupled to an end of the third pipe **800** proximate to the second pipe **104**. If a second pipe fitting were to be applied to an end of the second pipe, it would not be possible to fit another section of pipe in and still maintain the required pipe overlap.
(51) In a third stage shown in FIG. **10**, a left end of the first pipe **102**, which is the replacement segment of pipe joining the pipeline, is slid into the standard pipe fitting coupled to the third pipe **800**. The first pipe **100** may be rotated into place and then slid leftward to form the joint, as shown in FIG. **10**.
(52) In a final stage shown in FIG. **11**, the first pipe **102** has been slid such that an end of the first pipe **102** abuts the end of the third pipe **800**, and the longitudinal axes of all pipe portions **102**, **104**, **800** are aligned. The gap **110** now exists between the first pipe **102** and the second pipe **104**, and is spanned by the pipe coupling **100** installed as previously described, whereby the pipeline is joined.
(53) While the invention herein disclosed has been described by means of specific embodiments, examples and applications thereof, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope of the invention set forth in the claims.

Claims

1. A device for forming a tubular coupling, comprising: a fitting section having a longitudinal length and a generally half-tubular cross-section having an outside radius, wherein along the longitudinal length the fitting section comprises a left portion having a left inside radius and a left thickness, a right portion having a right inside radius and a right thickness, and a middle portion interposed between the left portion and the right portion and having a middle inside radius and a middle thickness, the fitting section further comprising: a first end of the cross-section having a receiving shape having a length, the receiving shape having an interior surface having the inside radius of the corresponding portion for the entire length of the receiving shape, and an exterior surface including a receiving shape locking face at a receiving distance from a terminus of the first end, wherein the interior surface and the exterior surface define a thickness of the receiving shape, wherein the receiving shape locking face comprises a first portion proximate to the terminus of the first end and a second portion contiguous with the first portion, wherein the second portion has a thickness less than the thickness of the first portion and the thickness of the second portion is at least 50% of the corresponding fitting section thickness; and a second end of the cross-section having a locking shape, the locking shape having a portion of an interior surface of the locking shape located outwards of the inside radius of the corresponding portion, thereby forming a notch in the interior surface of the locking shape, the locking flange notch including: a distal locking face located at a first distance from a terminus of the second end; and a proximate locking face located between the distal locking face and the terminus of the second end, whereby the notch has a width between the proximate locking face and the distal locking face, wherein the width of the notch is nominally equal to the receiving distance of the receiving shape; wherein the notch is configured to fit over and couple to a receiving shape of a second fitting section having the same geometry as the fitting section, including the receiving shape locking face of the second fitting section juxtaposed with the proximate locking face and the terminus of the first end of the second fitting section being located within the notch; and wherein the receiving shape is configured to receive and couple to a locking shape of the second fitting section including the receiving shape locking face juxtaposed with a proximate locking face of the second fitting section and the terminus of the first end being located within a notch of the second fitting section, whereby coupling of the fitting section to the second fitting section forms the tubular coupling, said tubular coupling having a tubular left portion having a tubular left inside radius, a tubular right portion having a tubular right inside radius, and a tubular middle portion having a tubular middle inside radius.
2. The device for forming the tubular coupling of claim 1, wherein the middle inside radius is

smaller than each of the left inside radius and the right inside radius.
3. The device for forming the tubular coupling of claim 2, wherein the left inside radius and the right inside radius are equal.
4. The device for forming the tubular coupling of claim 1, further comprising a waterproof adhesive applied to a portion of an interior surface of the fitting section.
5. The device for forming the tubular coupling of claim 1, wherein the fitting section is a material selected from the group consisting of PVC, CPVC, and ABS.
6. The device for forming the tubular coupling of claim 1, wherein the fitting section is formed by injection molding.
7. The device for forming the tubular coupling of claim 1, the second end further comprising a tooth, wherein the proximate locking face is included in the tooth.
8. The device for forming the tubular coupling of claim 1, wherein the receiving shape is configured to snap-lock couple to the locking shape of the second fitting section and the locking shape is configured to snap-lock couple to the receiving shape of the second fitting section.
9. The device for forming the tubular coupling of claim 1, wherein the locking shape has an exterior surface located outside the outside radius.
